



BUSINESS ANALYSIS

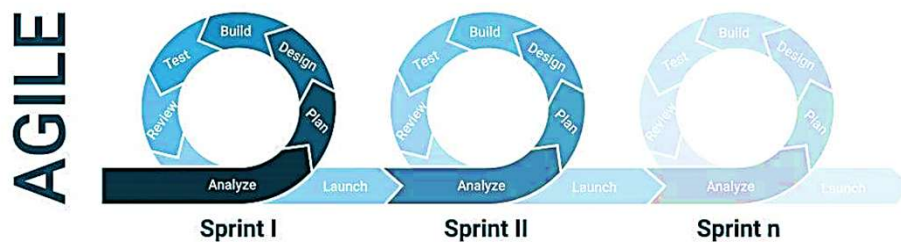
Agile, Scrum Roles & Backlog

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AGILE OVERVIEW

- **What is Agile Methodology:** A project management framework that breaks projects down into several dynamic phases, known as sprints.
- **Why use Agile:** To accommodate shifting needs and priorities of customers
- **How it works:** An iterative process in which teams set goals for each sprint, then build, test, and review their work with stakeholders before moving to the next sprint.



Source: asana; <https://asana.com/resources/agile-methodology>
 IT Chronicles: <https://itchronicles.com/agile/agile-methodology-what-it-means/>

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AGILE CONCEPTS

- **"Sprint":** A short work cycle, usually one to four weeks, where teams commit to completing specific tasks.
- **Product backlog:** A prioritized list of features, fixes, and tasks that guide what the team will work on next.
- **Sprint backlog:** The selected items from the product backlog (prioritized list of functions) that the team plans to finish.
- **Standup:** A short daily meeting where team members share progress, plans, and roadblocks.
- **Iteration:** A repeated cycle of planning, building, testing, and reviewing work to improve outcomes.
- **User story:** A brief explanation of a feature from the end user's perspective, often used to define requirements.
- **Epic:** A large work item made up of multiple user stories that span several sprints.
- **Velocity:** A metric showing how much work a team completes in a sprint, often measured in story points.
- **Burndown chart:** A visual tool that tracks remaining work against the time left in a sprint or project.
- **Work in progress (WIP):** The tasks currently being worked on; limiting WIP helps prevent bottlenecks and delays."

Source: asana; <https://asana.com/resources/agile-methodology>

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FOUR PILLARS OF AGILE

- **Individuals over processes and tools.** Agile teams value team collaboration and teamwork over working independently
- **Working software over comprehensive documentation.** The software that Agile teams develop should work; additional work, like documentation, is not as important
- **Customer collaboration over contract negotiation.** Customers are extremely important within the Agile methodology; customer collaboration is more important than the finer details of contract negotiation
- **Responding to change over following a plan.** Agile project management is very flexible; allows teams to quickly shift strategies and workflows without derailing an entire project.

Source: asana; <https://asana.com/resources/agile-methodology>

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DIFFERENT TYPES OF AGILE METHODOLOGIES

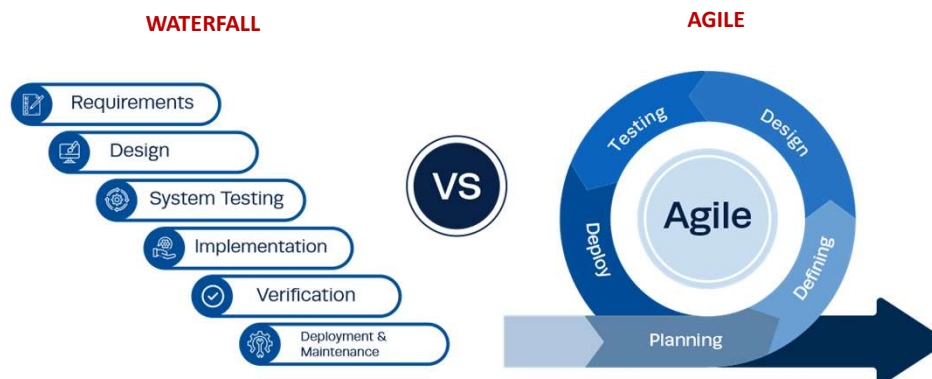
- **Kanban** is a visual approach to Agile; teams use online Kanban boards to show work in progress as tasks move through each stage of development.
- **Scrum** is a popular Agile methodology used by small teams and also involves sprints.
 - The team is led by a Scrum master whose primary role is to remove obstacles so team members can focus on their daily tasks.
 - Scrum teams hold daily meetings to discuss ongoing tasks, roadblocks, and other issues that might impact the development process.
- **Extreme Programming (XP)** emphasizes team values to improve collaboration. It has 5 core values that guide how developers interact and make decisions throughout the project:
 - Communication
 - Simplicity
 - Feedback
 - Courage
 - Respect
- **The Adaptive Project Framework** acknowledges that unknown factors can emerge and budgets, timelines, and team composition may have to be modified

Source: asana; <https://asana.com/resources/agile-methodology>

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AGILE VS. WATERFALL METHODOLOGY



Source: Institute Project Management; <https://instituteprojectmanagement.com/blog/waterfall-methodology/>

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SCRUM ROLES

- **The product owner** maximizes the value the product creates; defines the why, who, and what—why it is worthwhile to develop a product, who it is for, and what features it should contain.
 - Set product goals
 - Ensure that the product backlog is stocked with clearly expressed items that help reach the product goal
 - Order the backlog
 - Regularly update and refine the backlog together with the developers
 - Collect feedback from users, customers, and stakeholders on the product increments
 - Agree on sprint goals with the developers
- **The development team** creates products with the right quality; comprises a small team (2 to 3 people) that is:
 - Self-organizing
 - Cross-functional
- **The scrum master** coaches the product owner, developers, and stakeholders; he acts as a
 - Coach: Facilitate meetings, conversations, and improvements.
 - Protector: Run interference so the team can remain focused.
 - Leader: Lead without authority and puts the team first.
 - Advocate: Reinforce agile principles throughout the organization.

Source: SCRUMALLIANCE; <https://resources.scrumalliance.org/Article/scrum-team>

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BACKLOG

- **What is a Backlog/Product Backlog in Agile?**
A list that compiles all the tasks and user stories that must be done to complete the whole project
The sprint backlog is a subset of the product backlog and contains items that can be completed during each agile sprint.
- **What should be included in a Product Backlog?**
 - **User Stories & Epics:** Descriptions of a feature or functionality from the end user's perspective, e.g., "As a [user], I want [feature] so that [benefit]." Epics are larger user stories that require multiple sprints or tasks to complete and are later broken down into smaller, actionable user stories
 - **Technical Improvements:** Backend or infrastructure-related enhancements that improve system performance, scalability, security or maintainability. Examples include optimizing database queries or upgrading frameworks.
 - **Spikes** are time-boxed investigations or experiments to gain knowledge, reduce uncertainty or evaluate technical feasibility. They are often used when teams need to explore new technologies, architectural decisions or complex requirements before committing to implementation.
 - **Non-Functional Requirements:** These define system attributes such as performance, reliability, scalability and security.
 - **Dependencies & Constraints:** Dependencies are tasks that rely on completing other work items before they can be implemented. Constraints refer to limitations such as budget, time, regulatory requirements or technical restrictions that influence how a product is developed.
 - **Story Points:** Story points are a unit of measure used to estimate the effort required to complete a task or user story. They consider complexity, risk, and time, helping teams gauge workload and prioritize tasks without relying on exact time estimates.
 - **Acceptance Criteria:** The specific conditions and requirements that a product or feature must meet for it to be considered acceptable by stakeholders.
 - **Backlog Items Priority Levels and Status:** Backlog items' priority levels and status help teams organize and track work in an agile backlog. Priority levels (e.g., high, medium, low) define importance, while status (e.g., to-do, in progress, done) indicates progress toward completion.

Source: ProjectManager; <https://www.projectmanager.com/blog/product-backlog-sprint-backlog>

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BACKLOG EXAMPLE

Backlog Item ID	Type of Backlog Item	Description	Priority	Story Points	Status	Sprint	Assignee	Acceptance Criteria
1	Epic	User account management improvements	High	20	To Do	Sprint 1	Brenna	Users can create, delete, and reset accounts.
2	User Story	As a user, I want to reset my password so that I can regain access to my account	High	5	To Do	Sprint 1	Peter	User can reset password via email link.
3	Bug Fix	Fix login timeout issue	High	3	In Progress	Sprint 1	Bill	Login timeout no longer occurs after 5 minutes.
4	Technical Improvement	Refactor database queries for better performance	Medium	8	To Do	Sprint 1	Camilo	Query response time reduced by 30%.
5	Spike	Research feasibility of multi-factor authentication	Low	5	Completed	Sprint 1	Jorge	Report summarizing pros and cons of MFA.
6	Non-Functional Requirement	Improve page load speed to under 2 seconds	Medium	8	To Do	Sprint 1	Kaylie	Page load time under 2 seconds on average.
7	User Story	As a user, I want to reset my password via email so that I can regain access to my account.	High	5	In Progress	Sprint 2	Kaylie	User can reset password via email link.
8	User Story	Implement the "Forgot Password" functionality	High	3	In Progress	Sprint 2	John	"Forgot Password" button works and sends an email.
9	User Story	Design the password reset email template	Medium	2	To Do	Sprint 2	Sarah	Email template matches branding and includes reset link.
10	User Story	Integrate email service for password reset	High	3	To Do	Sprint 2	Mike	Reset email is sent successfully using the email service.
11	Bug Fix	Fix the checkout button not responding on mobile devices	Critical	2	Done	Sprint 2	Emma	Checkout button works on mobile devices.

Source: ProjectManager; <https://www.projectmanager.com/blog/product-backlog-sprint-backlog>

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SPRINT BACKLOG EXAMPLE

Backlog Item ID	Type of Backlog Item	Description	Priority	Story Points	Status	Sprint	Assignee	Acceptance Criteria
7	User Story	As a user, I want to reset my password via email so that I can regain access to my account.	High	5	In Progress	Sprint 2	Kaylie	User can reset password via email link.
8	User Story	Implement the "Forgot Password" functionality	High	3	In Progress	Sprint 2	John	"Forgot Password" button works and sends an email.
9	User Story	Design the password reset email template	Medium	2	To Do	Sprint 2	Sarah	Email template matches branding and includes reset link.
10	User Story	Integrate email service for password reset	High	3	To Do	Sprint 2	Mike	Reset email is sent successfully using the email service.
11	Bug Fix	Fix the checkout button not responding on mobile devices	Critical	2	Done	Sprint 2	Emma	Checkout button works on mobile devices.

Source: ProjectManager; <https://www.projectmanager.com/blog/product-backlog-sprint-backlog>

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PRODUCT BACKLOG VS. SPRINT BACKLOG

Feature	Product Backlog	Sprint Backlog
Purpose	Contains all potential work for the product	Defines work committed for the current sprint
Ownership	Managed by the Product Owner	Managed by the Development Team
Scope	Broad, covers the entire product	Focused, limited to a single sprint
Contents	Features, user stories, technical tasks, bug fixes, research items	Selected user stories, tasks, bug fixes, and sprint goals
Timeframe	Long-term, evolving over time	Short-term, updated only during the sprint
Prioritization	Continuously refined and reprioritized	Fixed during the sprint (unless critical changes occur)
Level of Detail	Can include high-level epics and partially defined tasks	Fully refined and broken down into actionable tasks
Flexibility	Items can be added, removed, or reprioritized at any time	Items are locked in once the sprint starts (except in special cases)

Source: ProjectManager; <https://www.projectmanager.com/blog/product-backlog-sprint-backlog>

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REAL WORLD EXAMPLES OF AGILE METHOD

• PayPal

- Problem: The company had become disorganized, bottlenecked, and slow to release new products.
- Strategy:
 - A team of 150 was assembled; coaches were engaged and formal training provided to thousands of employees.
 - A Scrum model was used and hundreds of cross-functional teams aligned around a two-week "sprint" (work) cycle.
- Results/Outcome: Within six months, the company had delivered 58 new products.

• Cisco

- Problem: The waterfall methodology was inefficient and created delays.
- Strategy: Cisco implemented SAgile for its Subscription Billing Platform as part of a broader shift to Agile methodologies
 - Cisco created three Agile Release Trains: Capabilities, Defects/Fixes, and Projects.
 - Teams coordinated for 15 minutes each morning to share progress and determine priorities, which enabled greater cohesion.
- Results/Outcome:
 - Workflows accelerated, overtime hours decreased, employees felt better overall.
 - There was 40 percent reduction in defects
 - There was 14 percent improvement in Defect Removal Efficiency.

Source: smartsheet; https://www.smartsheet.com/content/agile-project-management-examples?srsltid=AfmBOopMdRNNTKAQvCZIYrbx7yQp5P_D7eSP8tDyAUe3OEaNwdaFxN9s

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ACTIVITY

Imagine a hypothetical situation in which a business is facing declining profits and you are called upon to analyze the problem as a BA

- Describe the situation
- Provide descriptive statistics relevant to the situation, and perform
 - Diagnostic analysis
 - Predictive analysis
 - Prescriptive analysis
- Using the prescriptive analysis as a guide, describe the agile methodology you will use for your project to resolve the business problem and the sprint activities for four teams
- Last, describe four backlog items related to your project in the following table

Backlog Item No.	Type of Backlog item	Description	Priority	Story Points	Status	Sprint	Assignee	Acceptance Criteria

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ESSENTIAL READINGS

asana: "Manage your agile projects", available at <https://asana.com/uses/agile-management>

Institute Project Management: "What is waterfall methodology", available at <https://instituteprojectmanagement.com/blog/waterfall-methodology/>

IT Chronicles: "Agile methodology: What it means", available at <https://itchronicles.com/agile/agile-methodology-what-it-means/>

Project Manager: "Product backlog and sprint backlog management", available at <https://www.projectmanager.com/blog/product-backlog-sprint-backlog>

Scrum Alliance: "The scrum team roles and accountabilities", available at <https://resources.scrumalliance.org/Article/scrums-team>

smartsheet: "real-life agile project management & famous success stories", available at https://www.smartsheet.com/content/agile-project-management-examples?srsId=AfmBOopMdrNNTKAQvCZlYrbx7yQp5P_D7eSP8tDyAUe3OEaNwdaFxN9s

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REFERENCE BOOKS OPTIONS

- **BUSINESS ANALYSIS BODY OF KNOWLEDGE (BABOK)**; IIBA Bookstore: <https://www.iiba.org/career-resources/a-business-analysis-professionals-foundation-for-success/babok/>
- **BUSINESS ANALYSIS (Fourth Edition)** by Debra Paul (Author), James Cadle (Author), Malcolm Eva (Author), Craig Rollason (Author), Jonathan Hunsley (Author)
- **BUSINESS ANALYSIS TECHNIQUES: 123 ESSENTIAL TOOLS FOR SUCCESS (3RD Edition)** by James Cadle (Author), Debra Paul (Author), Jonathan Hunsley (Author), Adrian Reed (Author)